📘 Git & GitHub Ultra Cheat Sheet (Workflow Based)

🎯 Scenario: Upload a Project from Laptop to GitHub

Step 1 – Open Project Folder

```
cd D:\Projects\Calculator
```

📌 Go to your project folder.

Step 2 – Check Git Installation

```
git --version
```

✅ Verify Git is installed.

Step 3 – Initialize Git

```
git init
```

📌 Converts the project into a Git repository.

Step 4 – Check Status

```
git status
```

Output

```
Untracked files:
calculator.py
README.md
```

Step 5 – Add Files

Single file

```
git add calculator.py
```

All files

```
git add .
```

Step 6 – Verify

```
git status
```

Output

```
Changes to be committed:
calculator.py
README.md
```

Step 7 – First Commit

```
git commit -m "Initial Commit"
```

Step 8 – Create Repository on GitHub

Example

```
Repository Name
```

```
Calculator
```

Copy Repository URL

```
https://github.com/khushi/Calculator.git
```

Step 9 – Connect Local Project

```
git remote add origin https://github.com/khushi/Calculator.git
```

Step 10 – Verify Remote

```
git remote -v
```

Output

```
origin https://github.com/khushi/Calculator.git
```

Step 11 – Upload Project

```
git push -u origin main
```

✅ Project uploaded to GitHub.

🎯 Scenario 2 – Next Day You Added a New File

Created

```
login.py
```

Check

```
git status
```

Add

```
git add .
```

Commit

```
git commit -m "Added Login System"
```

Upload

```
git push
```

Done ✅

🎯 Scenario 3 – Download Latest Changes

Before starting work

```
git pull origin main
```

Now start coding.

🎯 Scenario 4 – Clone Existing Project

```
git clone https://github.com/khushi/Calculator.git
```

Project downloaded.

🎯 Scenario 5 – New Feature

Create branch

```
git branch feature-login
```

Switch

```
git checkout feature-login
```

OR

```
git checkout -b feature-login
```

After coding

```
git add .
git commit -m "Added Login Feature"
```

Switch

```
git checkout main
```

Merge

```
git merge feature-login
```

Push

```
git push
```

🎯 Scenario 6 – Wrong Code Written

Check changes

```
git diff
```

Undo

```
git restore calculator.py
```

🎯 Scenario 7 – Wrong File Added

Added

```
git add .
```

Oops!

Remove from staging

```
git reset login.py
```

🎯 Scenario 8 – View History

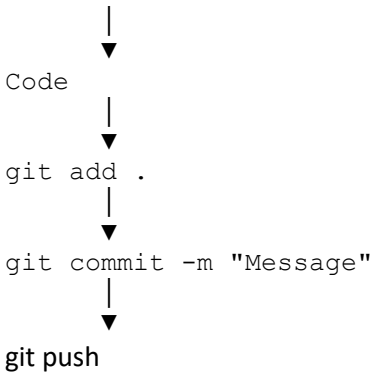
```
git log
```

Short version

```
git log --oneline
```

🚀 Complete Developer Workflow

```
Create Project
      |
      ▼
git init
      |
      ▼
git status
      |
      ▼
git add .
      |
      ▼
git commit -m "Initial Commit"
      |
      ▼
Create GitHub Repository
      |
      ▼
git remote add origin URL
      |
      ▼
git push -u origin main
      |
      ▼
Next Day
      |
      ▼
git pull
```


Code
git add .
git commit -m "Message"
git push